Combining Symbolic and Sub-symbolic Representations for Browsing a Large Document Repository

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Abstract

We propose a new document browsing method that combines symbolic and sub-symbolic representations of documents in a large repository; prove its feasibility with an implemented system, U-LISE, and apply it to a large scientific documents repository (SemOpenAlex). Our method consists in verbalizing the logical commonalities of two or more documents, that are selected by a user among the near ones in the embedding space. Such commonalities are verbalized as separate English phrases, that the user can vote for the relevance they attribute to the phrase. User's votes change the weights used in computing a new embedding, bringing closer embeddings that correspond to documents relevant for the search. In this way, we can support the sub-symbolic exploration in a completely interactive way, fully automatizing the loop between exploration and examination of results.

Keywords

RDF Knowledge Graphs, Knowledge Graph Embeddings, Repository Browsing, Logic-based Explainability, Human-in-the-Loop

1. Introduction

The increasing size of scientific document repositories makes their effective exploration a challenging Information Retrieval problem. A user interested in a scientific topic is rarely looking for a single document only. Rather, the search process often develops as an exploration of a portion of the scientific literature: an initially retrieved publication suggests other relevant works, authors, research topics, venues, or related areas, which in turn can lead to further documents of interest. Supporting this exploratory process is particularly important when the information need is not completely specified in advance, and evolves as the user becomes aware of the content and structure of the repository.

Traditionally, the exploration of scientific document collections relies mainly on *symbolic* information associated with publications. Documents can be retrieved or browsed according to their authors, titles, publication venues, keywords, research topics, citations, or terms occurring in their textual content. In repositories represented as Knowledge Graphs, this information can be even richer: *e.g.*, authors can be related as both co-authors of a third author, etc. through explicitly represented relations. Such symbolic representations have an important advantage for the user: the reason why a document satisfies a query, or why two documents appear to be related, can in principle be traced back to identifiable pieces of information.

In recent years, this form of exploration has been complemented by a different, *sub-symbolic* representation of documents. Machine Learning techniques can associate each document with a numerical vector, usually referred to as an *embedding*, in a high-dimensional space. Geometric proximity between embeddings can then be exploited as an additional criterion for document retrieval and repository browsing. Starting from a publication of interest, for example, the user can inspect publications whose embeddings are close to it, even when the relation between them cannot be captured by a simple equality of metadata values or by an explicit symbolic search condition.

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This possibility is particularly attractive for exploratory search. Embeddings may capture latent regularities emerging from the available information and can therefore suggest documents that would be difficult to retrieve through a previously formulated symbolic query. In this sense, sub-symbolic representations can complement conventional Information Retrieval mechanisms: instead of asking the user to state in advance exactly which properties a relevant document should satisfy, the system can propose documents that are somehow "similar" to those already considered interesting.

At the same time, this advantage introduces a problem. Geometric similarity does not directly explain itself. Once a document has been retrieved because its embedding is close to the embedding of another document, the user is still left with the question of *why* the two documents are close. Which properties or relations are responsible for their similarity? Do they concern their scientific topic, their methodological approach, some related research concepts, their authors or institutions, or other characteristics represented in the repository? More importantly from an Information Retrieval perspective, are the properties captured by the embedding actually *relevant to the user's current search*?

The latter question is especially important because relevance in an exploratory search is intrinsically user-dependent. Two publications may share several characteristics, but only some of them may be significant for the particular exploration being performed. For instance, similarity due to a common research area may be useful in one search session, whereas similarity based on some other metadata relation may be irrelevant. A numerical similarity value alone cannot allow the user to make this distinction, nor can it easily provide feedback about which aspects of similarity should influence the continuation of the search.

This paper proposes a new browsing process, in which both the symbolic and the sub-symbolic representation can be interleaved in one interactive loop. To substantiate our idea, we apply it to SemOpenAlex [1], a large repository of scientific publications whose descriptions are related through a large Linked Open Data Knowledge Graph. Such a symbolic graph representation provides explicit and inspectable information about publications. Our proposal is based on a closed-loop interaction between these two representations, that goes through the following steps (see Fig. 1).

1. We compute the embeddings of a selected subset of documents using the Knowledge Graph Embedding Model (KGEM) RDF2Vec [2], in its Python implementation, pyRDF2Vec [3]. These embeddings provide a geometric representation of publications that can be exploited to navigate among related resources. We stress that RDF2vec is a KGEM which uses weights for the predicates; initially, all weights are uniform, but they will be changed in the last step.
2. Users can browse the embedding space, and either select two or more publications that appear close to one another and are potentially relevant to the current exploration, or, alternatively, investigate the commonalities of collections of documents grouped together by a clustering process.
3. Rather than simply presenting the selected/clustered documents, we compute a logic-based symbolic characterization of the information they have in common, derived from the RDF representation of the selected resources.
4. Such commonalities are subsequently verbalized as separate natural-language phrases. The purpose is not merely to provide a plain English explanation of a similarity score, but to decompose the similarity among the selected documents into pieces of information (English phrases) that can be individually inspected by the user, who can vote their relevance from user's perspective.
5. The user's voting feedback is then used to modify the subsequent sub-symbolic representation. In particular, the relevance assigned by the user to the symbolic commonalities affects the weights employed when computing new embeddings. Properties regarded as more relevant can therefore have a stronger influence on the representation of the documents and consequently on their geometric proximity. Documents that are similar with respect to information considered important by the user should tend to become closer, while similarities based on information judged less relevant should have a smaller effect on the subsequent exploration. The loop can then restart from Step 1: computing embeddings, but with modified weights.

We prove the feasibility of our proposal in the context of SemOpenAlex, whose scale and rich graph structure make it a particularly significant case for this type of exploratory retrieval. SemOpenAlex contains more than 26 billion RDF triples and over 249 million scientific publications, together with a wide range of interconnected entities such as authors, institutions, venues, publishers, and research concepts[1]. Its heterogeneous and highly interconnected structure enables users to explore the literature along multiple semantic paths. In such a setting, traditional ranked lists provide only a partial view of the available information, as they present individual results without explicitly exposing the relationships among them or the alternative paths through which potentially relevant resources can be discovered. The availability of both a large amount of explicit semantic information about scholarly resources that can be represented through embeddings makes it possible to study how graph-based symbolic descriptions and their geometric representations can complement each other.

For example, a group of geometrically close publications may turn out to share information concerning a research concept, a related scientific field, or a more complex relation represented through paths in the Knowledge Graph. Since the common information is expressed through independent phrases, the user can indicate which of these commonalities are relevant to the current search and which are not. For instance, we show two RDF paths shared by a group of 155 works actually occurring in our experiments, together with one[1] of the possible explanations that could be provided by U-LISE (through a call to an LLM for English readability):

RDF: `_:x <https://semopenalex.org/ontology/hasConcept> _:z .`
`_:z skos:related <https://semopenalex.org/concept/C187736073> .`

LLM: *This group of resources is associated with a specific concept, which is related to the concept of "Management".*

RDF: `_:x <https://semopenalex.org/ontology/hasConcept> _:z .`
`_:z skos:related <https://semopenalex.org/concept/C199539241> .`

LLM: *This group of resources is associated with a specific concept, which is related to the concept of "Law".*

Note that the concept each resource is associated to varies (represented logically by a blank node, which is an existentially quantified variable), but that all these concepts, although different, are themselves related to "Management" (in the former case) or "Law" (in the latter).

Thus, symbolic information is used to make the results of a sub-symbolic representation inspectable, while user judgments over symbolic information are in turn used to influence the subsequent sub-symbolic representation. The two forms of representation are therefore not simply presented side by side, but cooperate during the browsing process.

In summary, we propose an interactive approach for browsing large document repositories that connects three operations which are usually treated separately: retrieving related resources through their embeddings, providing an explicit description of the information shared by retrieved resources, and exploiting user judgments about this shared information to influence recomputed embeddings. We present its implementation in U-LISE and illustrate the complete interaction cycle on scientific publications represented in SemOpenAlex.

The remainder of the paper is organized as follows. Section 2 discusses related approaches to embedding-based exploration of scientific document repositories and interactive/relevance-feedback methods. Section 3 describes U-LISE, while Section 4 presents its application to SemOpenAlex. Finally, the Conclusion discusses the current limitations of the approach and outlines future work. Appendices A–F (included just before References) provide the details to fully reproduce the results presented here (initial selection of resources in SemOpenAlex, RDF Stop patterns to be excluded, initial embedding through pyKEEN implementation of RDF 2vec, weights to be assigned to RDF 2vec for a second cycle).

[1]Since our plain English verbalization calls an LLM, the resulting phrases differ from one call to another, for the same RDF commonalities.

2. Related Work

The problem considered in this paper can be regarded as exploratory browsing: instead of issuing a new symbolic SPARQL query at each step, the user moves through a collection according to some representation of document similarity. Recent surveys on the related field of recommender systems show an increasing use of vector and graph representations, with contextual document embeddings becoming particularly important for content-based recommendation [4, 5]. The central issue for our work is not only whether two documents are close in an embedding space, but whether the system can also indicate *in what sense* they are similar in terms that a user can understand.

A prominent line of work linking exploratory browsing with embeddings learns vector representations of scientific papers from textual and citation information. SPECTER [6] learns document-level embeddings from title and abstract, using the (direct) citation graph as supervision, so that directly-citing/cited publications tend to be placed close in the learned space. SciNCL [7] refines this idea through neighborhood contrastive learning over a citation-graph embedding space, allowing graded relations among papers that can also be indirectly connected by citations. SPECTER2 [8] further provides task-adapted scientific document representations for settings including search and ranking. These methods offer effective representations for retrieving related papers, but the meaning of proximity remains primarily latent: once a nearby paper has been retrieved, the user is not given an explicit account of which concrete characteristics of the two papers make them similar. Distilling from the papers their commonalities (to see whether they are relevant for the specific search) is left to the user.

The *query-by-example* setting has also been studied directly in mainstream Information Retrieval. Zeng *et al.* [9] propose a personalized dense retrieval framework that unifies keyword search, query-by-example, and complementary item recommendation. In query-by-example, the content of an anchor item constitutes the request and is used to retrieve items that are nearby in the dense representations. Although their experiments concern e-commerce rather than scientific literature, the formulation is directly relevant to browsing items of a neighborhood in representation space. The retrieved neighborhood, however, is not accompanied by an explanation of the shared properties that justify the similarity between the anchor and each result.

A second line of research tries to make scientific-document similarity more controllable or interpretable. Ostendorff *et al.* formulate aspect-based document similarity [10], include the title of the section in which a citation occurs, as a label for the citation link. Such titles qualify the aspect-based similarity: methodology, background, or other facets. Their subsequent work on specialized document embeddings [11] represents papers through different embeddings for aspects such as task, method, and dataset. More recently, mCTRL [12] learns a generic representation together with several aspect-specific representations, enabling aspect-controllable related-paper recommendation. These approaches make the meaning of similarity less opaque, because the user can know or select the facet under which two papers are compared. Nevertheless, the aspect is predefined and comparatively coarse-grained: knowing that two papers are similar with respect to method is more gemeric than being told *which* method, property, or relation they actually share.

Readable representations have been used as part of the interaction loop. LIMEADE [13] allows users to inspect interpretable features behind recommendations and to provide positive or negative advice that affects the behavior of the underlying model. In its scientific-paper recommendation application, such features can include textual terms that influenced a recommendation. Closely related in spirit, Mysore *et al.* [14] introduce CtrlCE for controllable personalized scientific search. CtrlCE augments a cross-encoder with an editable memory constructed from a user's historical documents; it considers item-based and concept-based profiles, with the latter expressed through short natural-language concepts, and allows profile edits to influence subsequent rankings. However, CtrlCE tells the user what the system thinks the user's interests are; while our system U-LISE tells the user what the selected documents themselves have in common, taking into account users' votes on specific commonalities for progressively more refined search rounds.

This distinction is particularly relevant for *SemOpenAlex* [1], the repository considered in this work. SemOpenAlex represents scientific publications and related entities, including authors, institutions,

venues, related concepts, etc. as an RDF Knowledge Graph, SemOpenAlex also provides a Knowledge Graph embedding (using the Embedding Model DistMult [15]) for its entities. However, SemOpenAlex does not by itself explain why two particular publications happen to be close in the embedding space, nor the embedding model used can be tuned according to user's feedback.

Our proposed browsing method, implemented in the system U-LISE, connects the RDF symbolic level and a sub-symbolic embedding level during browsing, by using a Knowledge Graph Embedding Model (RDF 2vec) that allows for controlling through weights the sampling strategy used for computing the embeddings. Starting from of nearby publications selected in such embedding space, U-LISE computes commonalities from their RDF descriptions and verbalizes them as separate phrases in plain English. Hence, similarity is not explained only by a numerical score, a predefined aspect, or influential words, but by explicit properties and relations shared by the selected resources. Since these verbalized commonalities can be judged independently by the user, they can also provide relevance feedback for modifying the subsequent embedding-based exploration. In this way, U-LISE combines the ability of embeddings to expose latent neighborhoods with a symbolic account of similarity that can be inspected and understood by the user.

3. The system U-LISE

U-LISE has already been validated in its original version, LISE, on a different and larger dataset, with a deeper graph exploration depth [16, 17]. With respect to the classification of Semantic systems [18] based on RDF, U-LISE is on Level II regarding Blank Nodes (middle: treats them, without considering different instantiations), Level II regarding Entailment (middle: includes Simple Entailment, but it does not use full RDFS reasoning), and Level IV (topmost: plain English) regarding its Explanation facility.

In this work, together with the original automated clustering module, U-LISE embeds an alternative user-driven interaction mechanism: users can directly select the resources for which they wish to compute and visualize common information. Clustering and user selection represent two alternative interaction modules, depending on whether the goal is to analyze the entire dataset or a smaller subset of resources. These modifications highlight two key aspects of our system: (i) its ability to operate on datasets beyond those originally used, and (ii) its modular architecture, which allows individual components — such as clustering — to be replaced or reconfigured without affecting the overall pipeline. The architecture of U-LISE is shown in Fig. 1. In the next section we also report the results obtained through the original clustering module, in order to show how the system works in its original configuration and to highlight how the distribution of embeddings changes after their refinement based on user feedback. We refer to the initial execution of the U-LISE pipeline as the *first cycle*, and to the subsequent execution incorporating user-defined weights into the embedding process as the *second cycle*.

4. Application: browsing SemOpenAlex with U-LISE

We provide a practical demonstration of our idea by using our system U-LISE. Since in this paper we want just prove the feasibility of our approach, we do not conduct an extensive experimental evaluation, which we postpone to further research.

The dataset used in this work is SemOpenAlex [1], a large-scale (≈26B triples) scholarly Knowledge Graph derived from OpenAlex[2] and published in RDF format. In particular, a controlled subset is selected: the following data partitions are fully included from the official SemOpenAlex repository[3]: *topics*, *concepts*, *fields*, *subfields*, *keywords*, and *domains*, together with a subset of works from the partition *works_updated_date=2024-09-17_part_000*. From this partition, 50 resources of type *Work* associated with the concept *Artificial Intelligence* are extracted. The selection is performed through a SPARQL query (Appendix A) filtering resources by type and concept.

[2] https://openalex.org/

[3] https://semopenalex.s3.amazonaws.com/browse.html

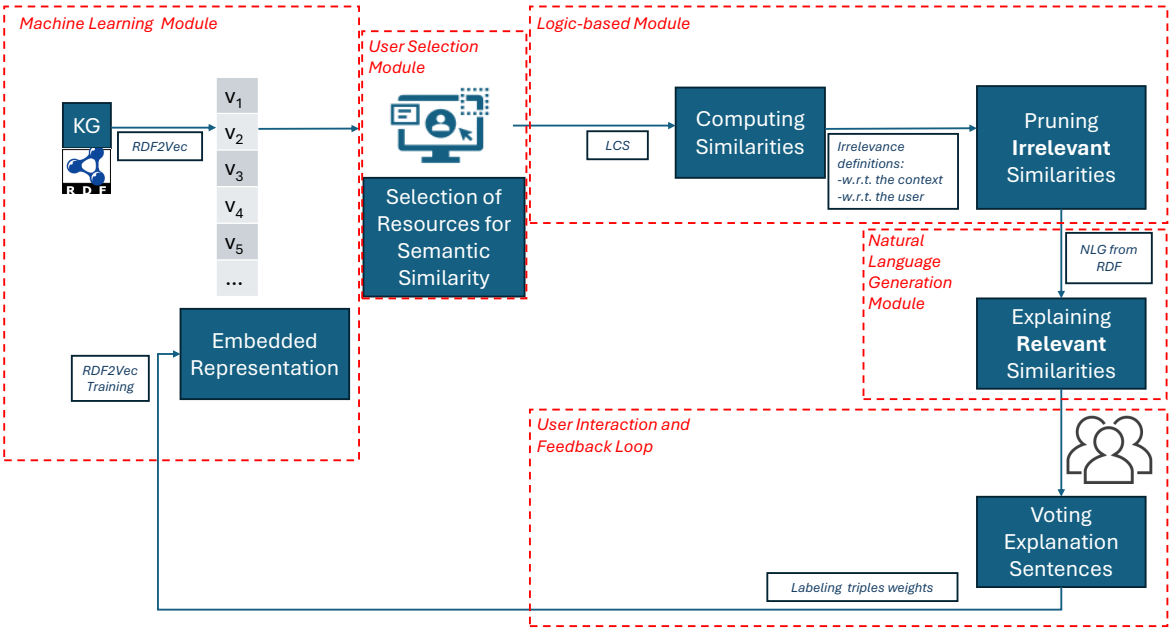


Figure 1: U-LISE architecture, highlighting the loop formed by the embedding exploration, verbalization of the commonalities of a chosen neighborhood of document embedding, user's vote and back for a refined embedding.

4.1. Step 1: Computing embeddings with a Knowledge Graph Embedding Model

pyRDF2Vec [3] is employed as KGEM due to its focus on capturing semantic and structural information from RDF graphs as well as its support for customizable embedding sampling strategies. Embeddings are learned with maximum 100 random walks per resource, maximum depth 3, and random seed 42. The sampler is Uniform for the *first cycle* and Predicate Relevance Weight [16] for the *second cycle* (the resulting weights assigned to each predicate are reported in Appendix B). All the other hyperparameters are kept at their default values. The distribution of learned embeddings in a three-dimensional space, obtained through Principal Component Analysis (PCA)[4], is shown in Fig. 2 (a) for the *first cycle* and in Fig. 2 (c) for the *second cycle*.

4.2. Step 2: User Selection of Resources through their Embeddings

Option 1: User Selection. The user selection of resources is enabled in two different ways:

1. through the interaction with a two-dimensional PCA visualization of the embeddings, in order to select geometrically near resources (enclosed in a user-defined box).
2. through the selection of resources from a list.

An example of the first selection method is shown in Fig. 3. The figure illustrates the selection of five resources, listed in Appendix C, directly from the projected embedding space.

Option 2: Clustering. Clustering is performed using k-Means[5]. The number of clusters k is determined via the elbow method over the range 2-15, with automatic detection by the *kneed* library[6], yielding $k = 8$ for the *first cycle* and $k = 7$ for the *second cycle*. We use k-Means++ initialization, 5 initializations (n_init), and random seed 42; all other parameters are kept at default values. Clustering is performed using the k-Means[7] algorithm. The optimal number of clusters k is determined via the

[4]https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html
[5]https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html
[6]https://pypi.org/project/kneed/
[7]https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html

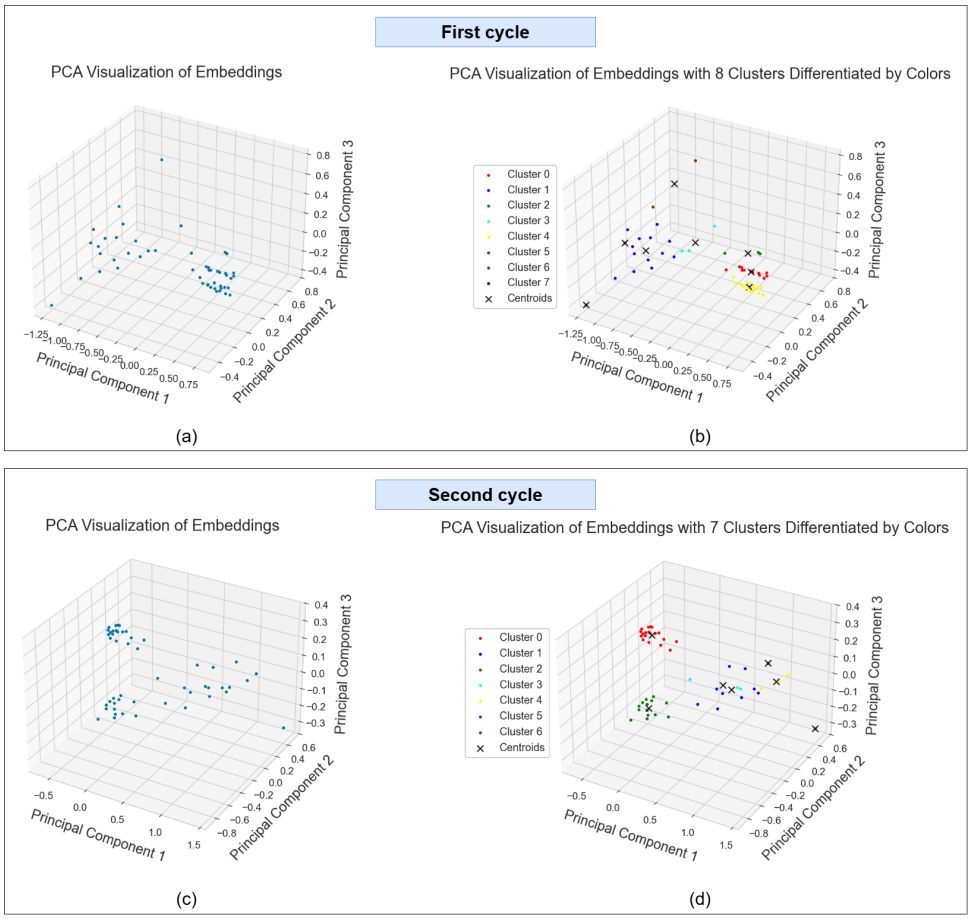


Figure 2: (a) the 3D-PCA projection of the embeddings (using Uniform Sampler) of the 50 resources selected in the *first cycle*; (b) embeddings in (a) partitioned into clusters (8 from the elbow method); (c) *second cycle*: the 3D-PCA projection of the new embeddings (using Predicate Relevance Weight Sampler) of the same 50 resources; (d) embeddings in (c) partitioned into 7 clusters, according to the elbow method.

elbow method over the range 2-15, with automatic detection by the *kneed* library[8]. This results in $k = 8$ for the *first cycle* and $k = 7$ for the *second cycle*. The clustering uses k-means++ initialization, 5 initializations (n_init), and random state 42; all other parameters are kept at default values. Results are shown in Fig. 2: before (b) and after (d) the embedding refinement, respectively.

4.3. Step 3: Logic-based Computation of RDF Commonalities

The Common Subsumer (CS) [19] is computed for the resources previously selected by the user (Sect. 4.2). During the computation, the *stop patterns* (*i.e.*, triple patterns considered irrelevant for similarity explanation) [16] are removed. The complete list of stop patterns is reported in Appendix D. The complete CS contains approximately 300 triples and is not reported due to space constraints, however, a few examples, verbalized in natural language, can be explored through the graphical interface presented in Sect. 4.5. Detailed examples of complete and pruned CSs can be found in [16].

4.4. Step 4: Natural Language Generation Module

RDF paths extracted from the CS are verbalized into natural language using the LLM *Gemini 3 flash*[9] with its default settings. Each path, together with its root node, is provided as input to the model

[8] https://pypi.org/project/kneed/
[9] https://docs.cloud.google.com/vertex-ai/generative-ai/docs/models/gemini/3-flash?hl=it

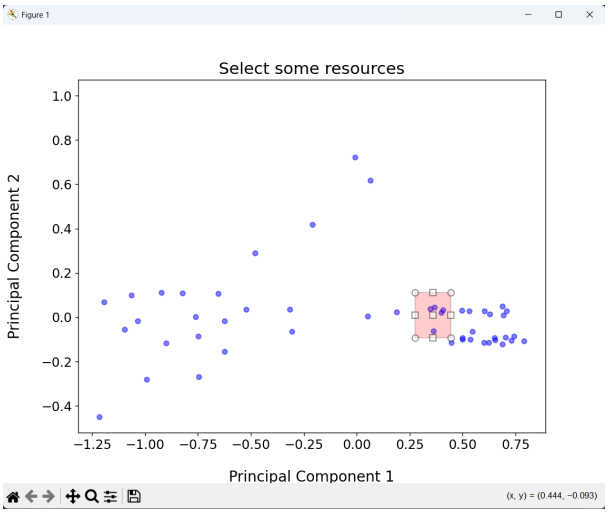


Figure 3: The interface which allows the user to select a group of resources in the 2D-PCA projection of the embedding space, in order to analyze the commonalities they share.

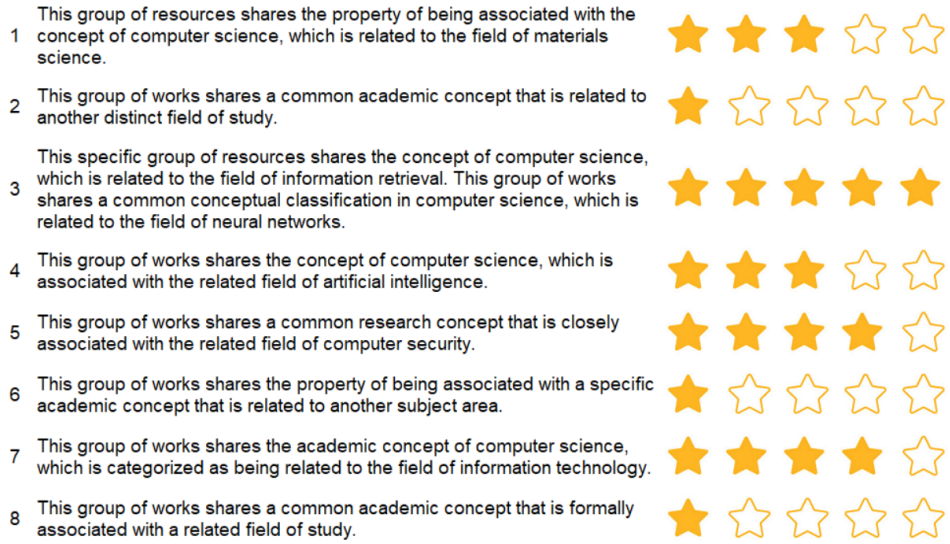


Figure 4: GUI allowing the user to weigh, piecewise, the relevance of common information shared by previously selected resources.

and transformed into a human-readable sentence via a structured prompt (Appendix E). Examples of generated explanations are shown in Fig. 4.

4.5. Step 5: User Interface and Feedback Loop

The generated sentences are presented through a Graphical User Interface (GUI). Fig. 4 displays a subset of the explanations derived from the CS computed over the five selected resources (Fig. 3). Users can rate the relevance of the shared information, and the resulting feedback is used to refine the embeddings in the second cycle, as illustrated in Fig. 2 (c)–(d).

5. Conclusion and Future Work

We proposed a fully interactive approach for browsing large document repositories that combines symbolic and sub-symbolic representations, and showed its feasibility by implementing it in our system U-LISE, and applying it to a large scientific document repository, SemOpenAlex. U-LISE enhances

the sub-symbolic clusterization in large Knowledge Graphs by: *(i)* computing the common logical characteristics of a given cluster — either user-selected or generated automatically by the clustering algorithm — of Knowledge Graph resources; *(ii)* verbalizing such commonalities in plain English; *(iii)* allowing users to specify piecewise which information is more relevant for them, and feed such preferences back into the pipeline of Vector Embedding → Resource Selection, enabling effective interactive refinement of the sub-symbolic representation. Currently, we are using U-LISE as a tool for assessing which combination of Knowledge Graph embedding model and Clustering algorithm provides the most explainable clusters in several domains — in particular, medical drugs.

Acknowledgments

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We provide below the details that allow researchers to fully reproduce of our results on SemOpenAlex.

A. SPARQL Query

```
SELECT DISTINCT ?work
WHERE {
  ?work rdf:type semopenalex_ontology:Work .
  ?work semopenalex_ontology:hasConcept semopenalex_concept:C154945302 .
}
LIMIT 50
```

The prefixes and namespaces used for this query, and throughout this work, are listed in Appendix F.

B. User-assigned Weights

Predicate	Weight
skos:related	2
dcterms:creator	4
semopenalex_ontology:crossrefType	2
semopenalex_ontology:hasAuthorship	8
semopenalex_ontology:isRetracted	2
semopenalex_ontology:oaStatus	2
semopenalex_ontology:metadataLanguage	2
semopenalex_ontology:hasConcept	8
semopenalex_ontology:workType	8

Table 1
Predicates with user-assigned weights for the Predicate Relevance Weight sampling strategy; all other predicates in the KG are assigned a weight of 1.

C. Selected Resources

```
https://semopenalex.org/work/W2224152705
https://semopenalex.org/work/W2274088303
https://semopenalex.org/work/W2302389139
https://semopenalex.org/work/W2554950137
https://semopenalex.org/work/W3200888896
```

D. Stop patterns

Stop patterns include all RDF triples $\langle s, p, o \rangle$ that meet at least one of the following conditions:

- the predicate p is one of the following:
  rdfs:seeAlso, owl:sameAs, semopenalex_ontology:umlsCui, semopenalex_ontology:imageThumbnail, foaf:depiction, prism:startingPage, prism:endingPage, semopenalex_ontology:countsByYear, semopenalex_ontology:citedByCount, dcterms:created, dcterms:modified, semopenalex_ontology:countriesDistinctCount, semopenalex_ontology:institutionsDistinctCount, semopenalex_ontology:worksCount, semopenalex_ontology:magId, semopenalex_ontology:level, dcterms:description, skos:note, semopenalex_ontology:locationsCount, datacite:doi, dcterms:abstract, skos:prefLabel, prism:publicationDate, skos:inScheme
- p is `rdf:type` and the object o is one of the following:
  skos:Concept, semopenalex_ontology:Concept, semopenalex_ontology:Topic, semopenalex_ontology:Subfield, semopenalex_ontology:Keyword, semopenalex_ontology:Authorship, semopenalex_ontology: OpenAccess ;

The above *stop patterns* are applied at depth levels 1 and 2. At depth level 3, the same *stop patterns* are applied and triples whose predicate is either skos:related or skos:broader are additionally excluded.

E. Prompt Template

Gemini 3 Flash is prompted with the following instruction:

> *Verbalize the following RDF path in natural English language:* ***RDF path***

The prompt template provided to the model is the following:

> *The following RDF path describes structured knowledge extracted from an RDF graph. Generate accurate, informative, and natural-sounding English sentences that clearly describe the meaning of the path. Each RDF path starts from the* ***root node*** *and represents a group of works sharing one or more properties. The RDF path provided contains one or more RDF triples referring to properties shared among works. The sentence must begin by making it clear that the properties are shared among that specific group of resources, and then state the shared property. Avoid vague or generic phrases like 'an unspecified resource' or 'an undefined item'; instead, infer a plausible general role or description. Produce exactly one coherent sentence per path. Avoid using URIs, internal identifiers (e.g., 'N...' or genid), or vague placeholders like 'reference', 'numerical identifier', or 'unspecified value'. Do not include RDF syntax, URIs, or technical metadata. Only provide clear, natural English language. If the URI is general, for example contains 'genid', treat it as a generic resource and describe its likely role meaningfully.*

The placeholders ***root node*** and ***RDF path*** are dynamically replaced with actual input values.

F. Prefixes and Namespaces Used

```
@prefix semopenalex_ontology: <https://semopenalex.org/ontology/>
@prefix semopenalex_concept: <https://semopenalex.org/concept/>
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#>
@prefix owl: <http://www.w3.org/2002/07/owl#>
@prefix foaf: <http://xmlns.com/foaf/0.1/>
@prefix prism:<http://prismstandard.org/namespaces/basic/2.0/>
@prefix dcterms: <http://purl.org/dc/terms/>
@prefix skos: <http://www.w3.org/2004/02/skos/core#>
@prefix datacite: <http://purl.org/spar/datacite/>
```

GenAI Usage Disclosure

Generative AI is used in our system for the natural-language verbalization of RDF Common Subsumers, as an external callable tool (Sec. 4.4). It has been used also to perform extensive search of references in main IR conferences, and only to generate outlines of Introduction and Related Work sections, which we used to write this manuscript.

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